Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

pastaisfine

Selected Domain

Domain 1: Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

SDG 8 (Decent Work and Economic Growth); secondary: SDG 4

Project / Solution Name

Roast My CV

Name (Leader)

Darren Tan Yik Wei

Email & Phone (Leader)

darrentyw5955@gmail.com

University Affiliation(s)

Universiti Teknologi Malaysia (UTM)

Number of Team Members

1

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian fresh graduates must choose a first career while AI changes which roles exist faster than curricula and career guidance can update. In 2024, 1.60 million graduates were employed below their qualifications, a skill-related underemployment rate of 32.2%. The country now has 5.98 million graduates. Across the workforce, 56% of professionals use AI tools at work but only 26% have received formal training (Hays Asia Salary Guide 2025), and the Ministry of Science, Technology and Innovation expects more than 600,000 workers to need reskilling within three to five years (both figures as cited in the Domain 1 brief)

Fresh graduates therefore face three gaps. They cannot tell which skills employers will still value. CV feedback is generic. Course lists are static and outdated when published. The consequences are wasted study years, lower earnings and slower national productivity, which engage SDG 4.4 and SDG 8.5. Many growing roles in health, clean energy and agriculture also advance other SDGs, but graduates rarely see that link when choosing a career. Our target user is a fresh graduate at the moment of choosing a first career.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Roast My CV lets a fresh graduate upload a CV (synthetic CVs only in the demo) and receive an honest, source-cited verdict on job readiness. An LLM Gemini 3.6 Flash extracts a structured skill profile and writes the “roast” in a tone the user selects: gentle, standard or blunt.

The trend signal layer ingests public job-posting data and DOSM demand data as dated snapshots. The interface displays each source and its date. Every occupation in MASCO major groups 1 to 5 carries an AI exposure score (0 to 10) with a written rationale. The Future Skills Radar is a treemap layered by demand, exposure and SDG.

Matching uses embeddings, computed on a RunPod endpoint, to compare the user’s skills with occupation skill profiles. The output is a skill-fit percentage and a skills-gap breakdown, with demand and AI exposure shown as separate labels. Two scenarios, moderate and fast AI disruption, re-weight exposure, so users see ranges rather than a single prediction.

The reskilling pathway is an ordered plan with estimated time, built from a short, manually verified list of free or low-cost providers. Each occupation and skill maps to at least one SDG through a table the AI drafts and I verify row by row.

Guardrails: every recommendation cites a source, the system cannot invent jobs, courses or statistics, and the user makes the final career decision. The interface supports English, Bahasa Malaysia and Chinese.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Stack: React (frontend); FastAPI and ExpressJs (backend); Gemini 3.6 Flash via Google AI Studio (extraction, roast text, batch re-scoring); embedding model and OCR Model on a RunPod endpoint; Replit (deployment); public job-posting and DOSM data.

Block diagram (text):
CV upload (OCR if necessary) → ExpressJs → LLM skill extraction → embedding (RunPod) → match against occupation skill profiles in codebase ← trend snapshots (dated, sourced) → exposure score and scenario engine → React UI (radar treemap, matches, gaps, pathway, SDG tags).
The ingestion job refreshes the snapshots, which triggers re-scoring.

Sponsor-provided resources: Yes. Replit accelerates my prototype development and RunPod serves the embeddings endpoint and hosting of my OCR Model

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: The persona is a fresh graduate choosing a first career, grounded in DOSM graduate statistics.
Define: The problem statement is that graduates cannot tell which skills remain valuable under AI disruption.
Ideate: Plain CV-chatbot designs were rejected in favour of a trend-grounded, scenario-based match.
Prototype: Build the single most important chain first (trend snapshot → exposure score → role match), then the pathway and SDG layers.
Test: Run synthetic profiles across all MASCO groups 1 to 5 and gather feedback from [number] peers on clarity and trust.
Deployment: The app is deployed early on Vercel with a public GitHub repo. The Phase 2 plan is to verify the course list, complete the SDG table review and add the Bahasa Malaysia and Chinese interfaces.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: by 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. Secondary: Target 8.5 (full and productive employment and decent work), listed in the Domain 1 brief.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	32.2% graduate skill-related underemployment in Malaysia in 2024 (1.60 million graduates employed below their qualifications). Source: DOSM Graduates Statistics 2024, via Bernama.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Skill-fit gain: the average percentage-point increase in a user's skill-fit score for their chosen target role, measured before and after the pathway's skills are completed. This tracks the skill mismatch behind the 32.2% graduate skill-related underemployment rate recorded in 2024. Supporting KPI: citation coverage, the percentage of recommendations (roles, courses, statistics) that carry a verifiable source and date
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed: 100% of recommendations cited and an average skill-fit gain of at least 20 percentage points on test profiles after completing the pathway.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code	MIT
ExpressJs	Code	MIT
Gemini 3.6 Flash	Model	Free to use
Embedding model (RunPod)	Model	MIT
OCR model (RunPod)	Model	Apache License 2.0

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

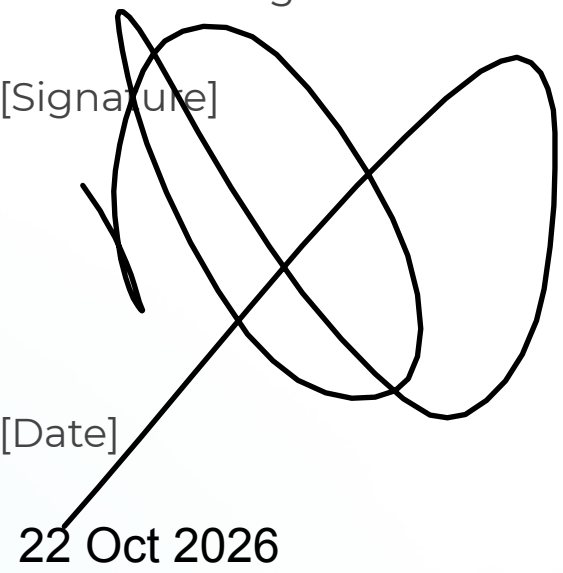
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

22 Oct 2026